

Uncertainty Visualization in Data Journalism: A Pilot Study on Election Polls in a Der Spiegel Online Article

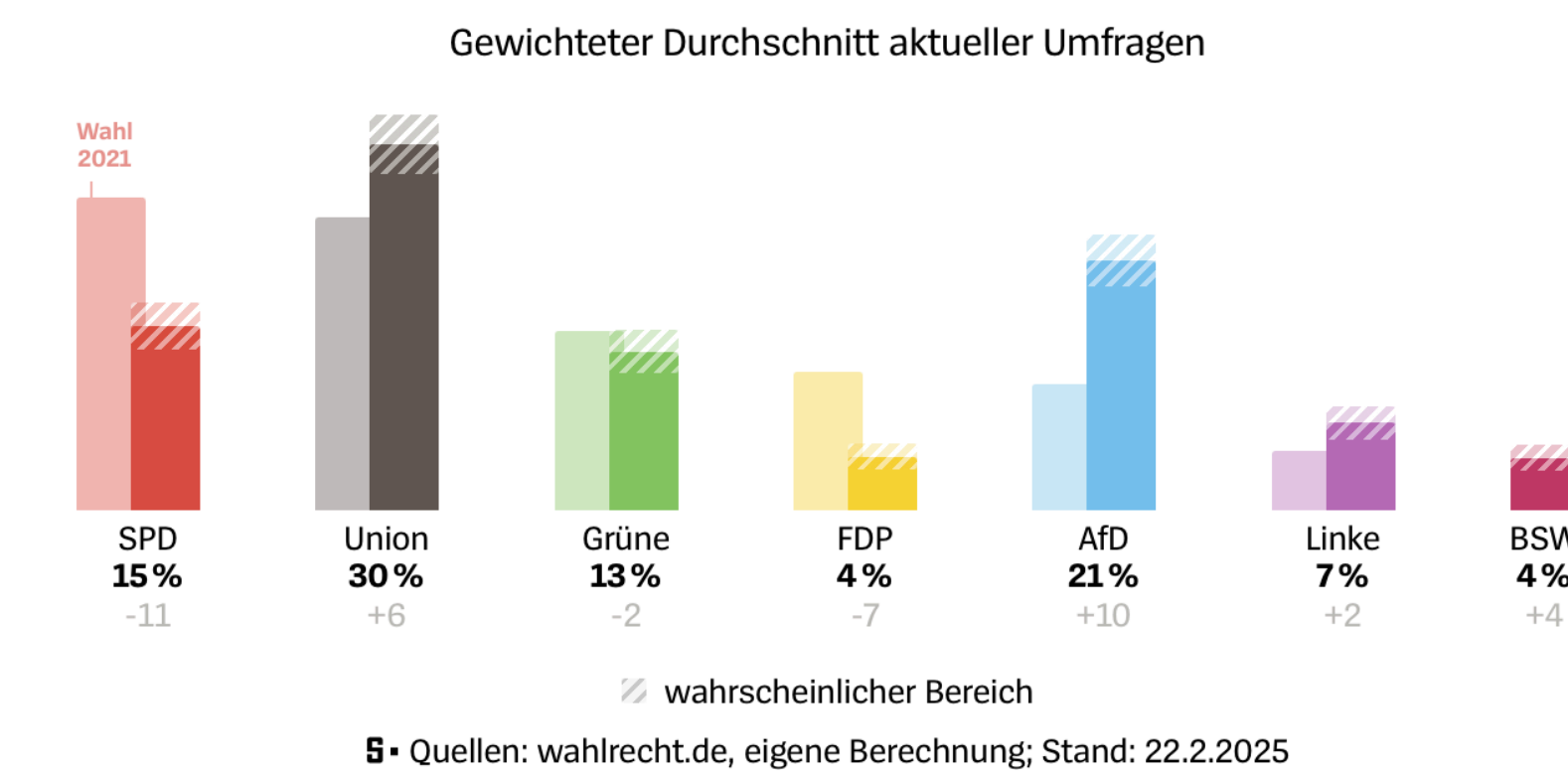
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1 Uncertainty in Journalism

- Uncertainty is key in polling - margins of error, sample variation.
- Lab studies exist, but little research on real-world readers.
- This study explores how Der Spiegel readers interpret uncertainty visualization in polls.



2 Research Questions

- How do readers perceive and interpret uncertainty visualizations?
- Which aspects are correctly understood vs. misinterpreted?
- How does this inform design for data journalism?

3 Methods

- Online survey embedded in Der Spiegel online article (Feb 17–23).
- N = 98 participants; mostly men (73%), broad age range, highly educated (62%).
- Participants rated usefulness of confidence intervals for three levels of engagement:
 - **Significance**: general comprehension.
 - **Analysis**: comparing party shares.
 - **Interpretation**: possible election outcomes, implications.
- Free-text responses for deeper insights.

4 Results

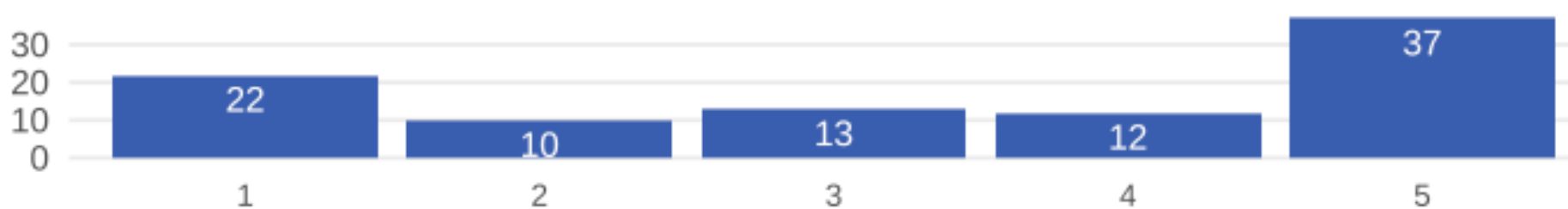
Overall Perceived Usefulness

"67% found the visualization useful"

Distributions of closed questions

"The representation of the probable range helps me to better..."

... assess the **Significance** of the results".

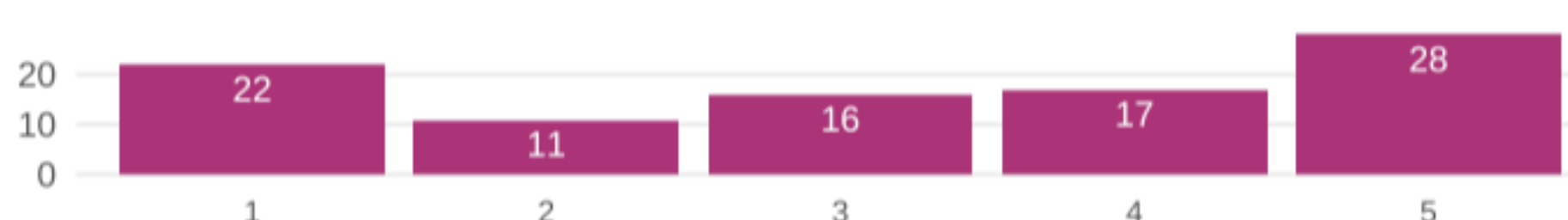


.... compare the predicted vote shares of the parties"

Analysis



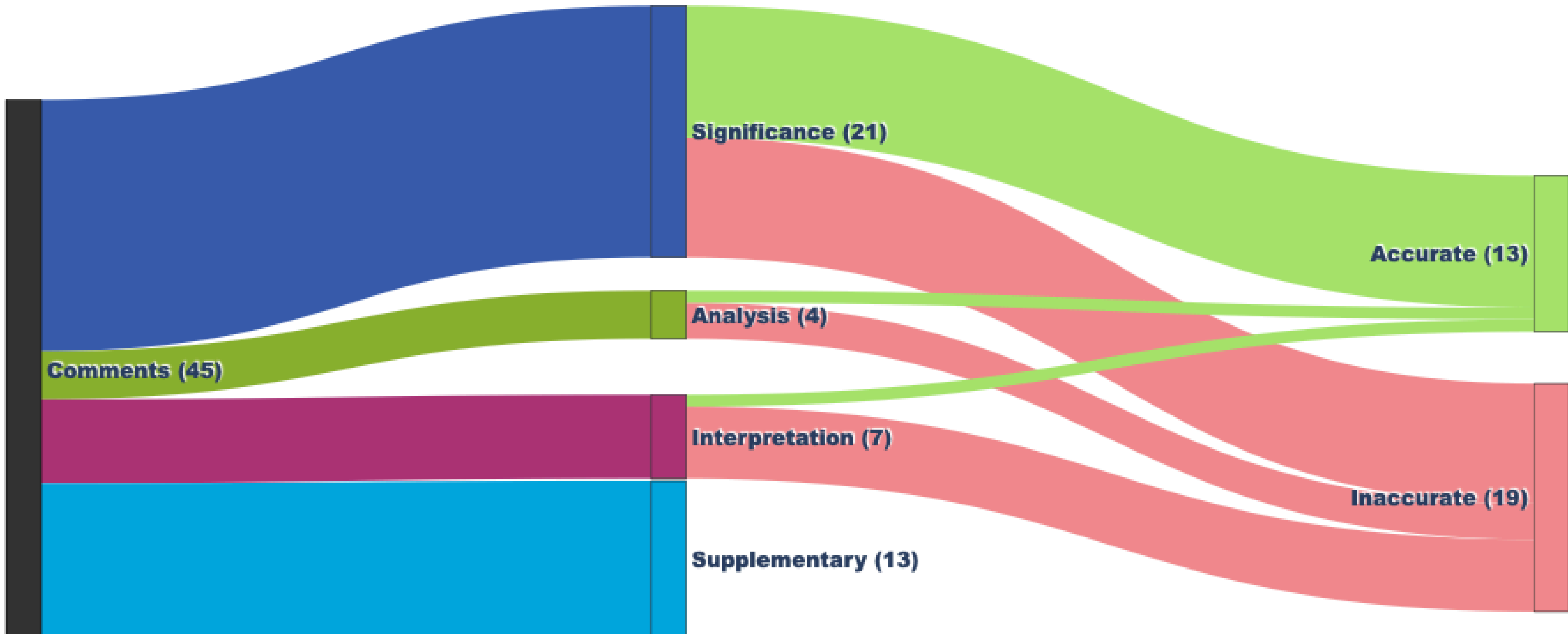
....assess possible election outcomes" **Interpretation**



1 = Strongly disagree, 5 = Strongly agree

Responders tend to polarize on these questions.

Distribution of open-ended questions:



Open ended responses

- 45 responses
- 13 accurate interpretations
- 19 inaccurate interpretations
- 13 supplementary remarks (comments, suggestions, critiques and opinions)

Examples comments:

Significance

Accurate:

"[...]It is clear anyway that there is an area of uncertainty"

Inaccurate:

"[...]distributing deviations of 1% everywhere is clumsy[...]"

Analysis

Accurate:

"Helps to assess, for example, whether the 5% hurdle is likely to be exceeded and, if so, whether it is rather certain or rather shaky"

Inaccurate:

"The correct order would be CDU AfD SPD Greens ... so that the conservative majority in the country (55%, and rising) would be honestly represented"

Interpretation

Accurate:

"I see the fuzziness of the forecasts more clearly. For example, the FDP may or may not be out. . ."

Interpretation

Inaccurate:

"[...]this only creates manipulation and the voter is lazy and does not read the election programmes [...]"

"To warn against voting for the FDP"

5 Learned remarks

- **Accurate:** Some readers understood error bands (robustness).
- **Inaccurate:** Others saw them as manipulation or confusion.
- **Attention bias:** Headline percentages overshadow uncertainty.
- **Trust factor:** Reader skepticism toward polls shapes interpretation.
- **Overall:** Visuals appreciated, but communication gap persists.

6 Limitations and future work

- Convenience sample, these are voluntary readers of the article → larger scale
- Reader trust & skepticism strongly shaped interpretations → trust building strategy
- Confidence intervals not intuitive for many readers → alternative representation

7 Concluding remarks

- Our pilot study confirms prior findings that confidence intervals are not intuitively understood.
- Readers varied in their interpretations, with some focusing solely on headline percentages, overlooking error bars.
- Attitudes toward polls and trust in the media influenced responses.
- Need for clearer, more guided designs for uncertainty.

The results suggest that future visualizations should more actively guide readers' attention to uncertainty, and additional studies should explore design strategies that reduce misinterpretation while maintaining credibility.